TrES-5b

R-0938

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_230612` · created 2026-07-29T12:26:23+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460107.8817 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1824 +/- 0.0083
Transit depth (Rp/Rs)^2	3.33 +/- 0.3 [%]
Orbital Inclination (inc)	84.62 +/- 1.45
Airmass coefficient 1 (a1)	1.107 +/- 0.015
Airmass coefficient 2 (a2)	-0.083 +/- 0.033
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#2 - [289, 358]
Optimal Method	PSF photometry
Transit Duration (day)	0.0782 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1824 ± 0.0083 vs 0.142 (deviation 4.87σ)
Duration fit vs published	0.0782 d vs 0.0758 d (deviation 0.35σ)
Mid-time O–C	-2.3 min
Depth SNR	22.0
Residual scatter	1.35 %

Light curve

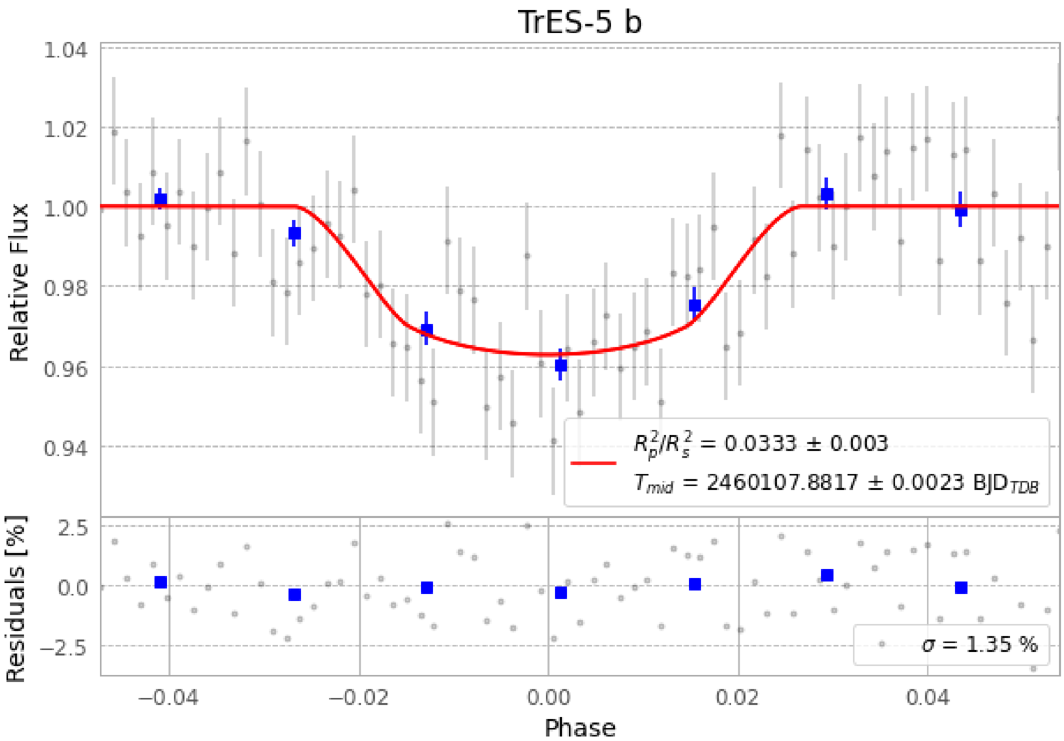

Figure 1 — FinalLightCurve_TrES-5 b_2023-06-12.png (SHA-256 c902f64db2e72195…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 220], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 55e8f982f5e6f213…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

73 FITS frames (48 MB), `TRES-5230612072715.FITS` → `TRES-5230612110315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230612.log (4b82763a82ad…), FinalLightCurve_TrES-5 b_2023-06-12.png (c902f64db2e7…), FinalLightCurve_TrES-5 b_2023-06-12.csv (02db7276f08f…)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-29 12:26Z.